

WBS Dictionary

Fitness Tracking and Meal Planning Mobile Application

GOAL

The goal of this project is to develop an Android mobile application that helps users improve their health habits by combining fitness activity tracking and meal planning in one platform. The app will provide clear progress summaries and dashboards to support goal monitoring, while also securely storing user data using Firebase services.

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1.0 Requirements Analysis

1.1 Functional Requirements Document (FRD)

Description: Complete list of functional features (tracking, meal planning, goals, reports, notifications)

Assigned: Business Analyst

Expected Output: Approved FRD (versioned PDF/Doc)

Estimated Timeline: 1–2 weeks

Tools/Tech: Google Docs, interviews, surveys

1.2 Non-Functional Requirements Document (NFRD)

Description: Performance, security, usability, availability, privacy requirements.

Assigned: BA + Tech Lead

Expected Output: Approved NFRD

Estimated Timeline: 3–5 days

Tools/Tech: Google Docs

1.3 User Stories Package

Description: User stories per role (user/admin) based on the required app features.

Assigned: BA / Product Owner

Expected Output: User stories backlog document/sheet

Estimated Timeline: 1 week

Tools/Tech: Google Sheets

1.4 Requirements Sign-off Package

Description: Final review notes and sign-off record of requirements baseline.

Assigned: BA + Project Manager

Expected Output: Signed approval page/minutes of meeting

Estimated Timeline: 1–2 days

Tools/Tech: Google Docs

1.5 Requirements Traceability Matrix (RTM)

Description: Mapping FRD/NFRD → design modules → test cases.

Assigned: QA Lead

Expected Output: RTM spreadsheet/document

Estimated Timeline: 2–4 days (then maintained)

Tools/Tech: Google Sheets

2.0 System Design

2.1 UX Research & User Flow Package

Description: Personas, journey maps, and user flows (onboarding, logging, meal plan, dashboard)

Assigned: UX Designer

Expected Output: User flow diagrams + UX notes

Estimated Timeline: 1 week

Tools/Tech: Figma

2.2 Wireframes & Interactive Prototype Package

Description: Wireframes + clickable prototype for main screens.

Assigned: UI/UX Designer

Expected Output: Prototype link + screen list

Estimated Timeline: 1–2 weeks

Tools/Tech: Figma

2.3 UI Style Guide Package

Description: Color palette, typography, components, spacing, icons.

Assigned: UI Designer

Expected Output: Style guide + reusable UI components

Estimated Timeline: 3–5 days

Tools/Tech: Figma design system

2.4 System Architecture Design Package

Description: Architecture diagram (mobile app + Firebase backend + integrations), security approach.

Assigned: Solution Architect/Tech Lead

Expected Output: Architecture document/diagram

Estimated Timeline: 3–7 days

Tools/Tech: Google Docs

2.5 Database Design Package (ERD + Data Dictionary)

Description: ERD for users, goals, workouts, meals, logs, nutrients, notifications.

Assigned: Backend Lead / DBA

Expected Output: ERD + data dictionary

Estimated Timeline: 3–7 days

Tools/Tech: dbdiagram.io, Firebase Console

2.6 Firebase Data Model & Security Rules Specification Package

Description: Defines Firestore collections/documents structure, Firebase Authentication flow, and Firestore Security Rules for data access.

Assigned: Backend Lead

Expected Output: Firebase specification document (Firestore data model + Security Rules overview)

Estimated Timeline: 1 week

Tools/Tech: Firebase Console, Google Docs

3.0 Development

3.1 Mobile Application (Frontend)

Description: React Native Android UI modules: auth, profile/goals, tracking, meals, macros, dashboard, notifications, settings.

Assigned: Mobile Developer (React Native)

Expected Output: Working Android mobile app frontend with all screens/modules connected to Firebase.

Estimated Timeline: Week 3–5

Tools/Tech: Google Docs / Figma

3.1.1 Authentication & Onboarding Module

Description: Login/register, onboarding steps, consent/privacy prompts.

Assigned: Mobile Developer

Expected Output: Working authentication and onboarding screens integrated with Firebase Authentication.

Estimated Timeline: 1–2 weeks

Tools/Tech: React Native CLI, JavaScript, VS Code, Android Studio, Git/GitHub, Firebase SDK

3.1.2 User Profile & Goals Module

Description: Profile edit, goal setup (weight, calorie, steps), preferences.

Assigned: Mobile Developer

Expected Output: Profile and goals screens with data saved to and retrieved from Cloud Firestore

Estimated Timeline: 1 week

Tools/Tech: React Native CLI, JavaScript, VS Code, Android Studio, Git/GitHub, Firebase SDK

3.1.3 Fitness Tracking Module (manual logging + summaries)

Description: Manual activity logging (steps/time/calories) and daily summary view.

Assigned: Mobile Developer

Expected Output: Fitness tracking UI with activity logs stored in Cloud Firestore and displayed in summaries

Estimated Timeline: 1–2 weeks

Tools/Tech: React Native CLI, Cloud Firestore (Firebase SDK), Git/GitHub

3.1.4 Workout Plan Module

Description: Workout plan list, workout details, completion logging.

Assigned: Mobile Developer

Expected Output: Workout plan screens with workout completion records saved to Cloud Firestore

Estimated Timeline: 1–2 weeks

Tools/Tech: React Native CLI, Cloud Firestore (Firebase SDK), Git/GitHub

3.1.5 Meal Planning Module

Description: Meal plan builder (breakfast/lunch/dinner/snacks), schedule.

Assigned: Mobile Developer

Expected Output: Meal planning UI with meal plans saved to and loaded from Cloud Firestore

Estimated Timeline: 1–2 weeks

Tools/Tech: React Native CLI, Cloud Firestore (Firebase SDK), Git/GitHub

3.1.6 Nutrition & Macro Tracking Module

Description: Food search/add, macros summary (calories, protein, carbs, fats)

Assigned: Mobile Developer

Expected Output: Food logging UI with macro totals computed and entries stored in Cloud Firestore

Estimated Timeline: 2 weeks

Tools/Tech: React Native CLI, Cloud Firestore (Firebase SDK), Git/GitHub

3.1.7 Progress Dashboard & Analytics Module

Description: Charts for weight trend, calories, workouts, adherence.

Assigned: Mobile Developer

Expected Output: Dashboard screens with charts generated from Cloud Firestore data

Estimated Timeline: 1–2 weeks

Tools/Tech: React Native CLI, react-native-chart-kit, Git/GitHub

3.1.8 Notifications & Reminders Module

Description: Reminders for meals/workouts/water (as defined), notification preferences.

Assigned: Mobile Developer

Expected Output: Push notifications working via Firebase Cloud Messaging (FCM) with reminder preference handling.

Estimated Timeline: 1 week

Tools/Tech: Firebase Cloud Messaging (FCM), React Native CLI

3.1.9 Settings & Preferences Module

Description: Units (kg/lb), privacy settings, data export request (if required), logout.

Assigned: Mobile Developer

Expected Output: Settings screens with preferences saved and applied within the app

Estimated Timeline: 3–5 days

Tools/Tech: React Native CLI, JavaScript, VS Code, Android Studio, Git/GitHub

3.2 Firebase Backend Services

3.2.1 Authentication & Authorization Configuration

Description: Configure user sign-up/sign-in and access control using Firebase Authentication.

Assigned: Backend Developer

Expected Output: Working Firebase Auth (registration/login) integrated with the mobile app

Estimated Timeline: 1–2 weeks

Tools/Tech: Firebase Authentication, Firebase Console

3.2.2 User Profile & Goals Data Service

Description: Set up Firestore data structure for user profiles, goals, and preferences.

Assigned: Backend Developer

Expected Output: Firestore collections for profiles/goals; create/read/update working in the app

Estimated Timeline: 1 week

Tools/Tech: Cloud Firestore, Firebase Console

3.2.3 Fitness Tracking Data Service

Description: Store and retrieve activity logs and workout completion records in Firestore.

Assigned: Backend Developer

Expected Output: Firestore collections for fitness logs; saving and viewing logs working

Estimated Timeline: 1–2 weeks

Tools/Tech: Cloud Firestore, Firebase Console

3.2.4 Workout Plan Data Service

Description: Store workout plans/templates and user workout schedules in Firestore.

Assigned: Backend Developer

Expected Output: Workout plan data stored in Firestore; app can load and update plans

Estimated Timeline: 1–2 weeks

Tools/Tech: Cloud Firestore, Firebase Console

3.2.5 Meal Planning & Nutrition Data Service

Description: Store meal plans, food logs, and nutrition/macro entries in Firestore.

Assigned: Backend Developer

Expected Output: Meal/nutrition data stored in Firestore; app can add/edit/view entries

Estimated Timeline: 2 weeks

Tools/Tech: Cloud Firestore, Firebase Console

3.2.6 Analytics/Progress Queries Package

Description: Create queries needed for dashboard summaries (daily/weekly progress).

Assigned: Backend Developer

Expected Output: Working Firestore queries used by dashboard screens

Estimated Timeline: 1 week

Tools/Tech: Cloud Firestore, Firebase Console

3.2.7 Push Notifications Configuration

Description: Configure push notifications for reminders and updates.

Assigned: Backend Developer

Expected Output: FCM configured; device token saving and test push notification working

Estimated Timeline: 1 week

Tools/Tech: Firebase Cloud Messaging (FCM), Firebase Console

3.2.8 Security Rules & Error Logging Setup

Description: Configure Firestore Security Rules and basic app error logging.

Assigned: Backend Developer / Tech Lead

Expected Output: Security rules applied; error logging enabled for troubleshooting

Estimated Timeline: 3–5 days

Tools/Tech: Firebase Security Rules, Firebase Console, Firebase Crashlytics

3.3 Cloud Firestore Database Implementation

3.3.1 Firestore Collections & Document Structure Package

Description: Implement the final Firestore collections/documents based on the database design.

Assigned: Backend Developer

Expected Output: Firestore database structure created and ready for app use

Estimated Timeline: 1 week

Tools/Tech: Cloud Firestore, Firebase Console

3.3.2 Seed/Reference Data Package

Description: Add initial/reference data (units, categories, default lists if needed).

Assigned: Backend Developer

Expected Output: Reference data available in Firestore

Estimated Timeline: 3–5 days

Tools/Tech: Cloud Firestore, Firebase Console

3.3.3 Backup & Restore Package

Description: Create backup/export procedure and restore steps for Firestore data.

Assigned: DevOps / Backend Developer

Expected Output: Backup/restore steps document and test export (if applicable)

Estimated Timeline: 2–3 days

Tools/Tech: Firebase CLI, Cloud Firestore

3.4 Integrations

3.4.1 Wearable/Health Data Integration (Google Fit)

Description: Read health metrics (steps, calories, workouts) based on permissions.

Assigned: Mobile Dev + Tech Lead

Expected Output: Integrated data import + permission handling

Estimated Timeline: 1–2 weeks

Tools/Tech: Google Fit API

3.4.2 Nutrition/Food Data API Integration

Description: Food search and nutrition facts via third-party API.

Assigned: Backend Dev (primary) + Mobile Dev (UI)

Expected Output: Working food search + nutrition detail retrieval

Estimated Timeline: 1–2 weeks

Tools/Tech: USDA FoodData Central API, Postman

3.4.3 Push Notifications Integration (FCM)

Description: Device token registration and push delivery setup.

Assigned: Mobile Dev + Backend Dev

Expected Output: Push notification end-to-end working

Estimated Timeline: 3–7 days

Tools/Tech: Firebase Cloud Messaging (FCM)

4.0 Testing

4.1 Test Plan Package

Description: Testing scope, strategy, test types, entry/exit criteria.

Assigned: QA Lead

Expected Output: Approved test plan

Estimated Timeline: 2–4 days

Tools/Tech: Google Docs, RTM (Google Sheets)

4.2 Unit Test Suite

Description: Unit tests for core app logic and utilities.

Assigned: Developers + QA oversight

Expected Output: Automated unit tests + report

Estimated Timeline: Ongoing (1–2 weeks initial)

Tools/Tech: Jest

4.3 Integration Test Suite

Description: Integration tests for Firebase Authentication/Firestore operations and data validation.

Assigned: QA Engineer

Expected Output: Integration test results

Estimated Timeline: 1 week

Tools/Tech: Postman

4.4 System & End-to-End Test Report

Description: End-to-end testing of user flows (log workout, plan meals, dashboard).

Assigned: QA Team

Expected Output: System test report

Estimated Timeline: 1–2 weeks

Tools/Tech: Manual testing, Android Emulator/Real device

4.5 Usability Test Report

Description: Usability feedback from test users; improvements list.

Assigned: UX Designer + QA

Expected Output: Usability report + recommended changes

Estimated Timeline: 3–5 days

Tools/Tech: Survey forms, observation notes

4.6 Security Test Report

Description: Basic security checks (Firebase Authentication, Firestore Security Rules, data protection).

Assigned: QA Lead / Tech Lead

Expected Output: Security test report

Estimated Timeline: 3–5 days

Tools/Tech: OWASP Mobile checklist

4.7 Performance Test Report

Description: App responsiveness and Firestore query performance checks.

Assigned: QA / Tech Lead

Expected Output: Performance results report

Estimated Timeline: 3–5 days

Tools/Tech: Postman

4.8 UAT Sign-off Package

Description: User acceptance testing scripts and sign-off.

Assigned: Product Owner/Client + QA Facilitator

Expected Output: UAT sign-off document

Estimated Timeline: 3–7 days

Tools/Tech: UAT checklist, forms

4.9 Defect Closure Report

Description: Defect summary, fixed/deferred list, final quality status.

Assigned: QA Lead

Expected Output: Defect closure report

Estimated Timeline: 2–3 days

Tools/Tech: Trello defect log

5.0 Deployment

5.1 Environment Setup Package (Dev/Test/Prod)

Description: Hosting, database, secrets/config, access controls.

Assigned: DevOps

Expected Output: Working environments + config docs

Estimated Timeline: 1 week

Tools/Tech: Firebase Project, Firebase Console

5.2 CI/CD Pipeline Package

Description: Automated build/test/deploy pipeline.

Assigned: DevOps

Expected Output: Pipeline config + build artifacts

Estimated Timeline: 1 week

Tools/Tech: GitHub Actions

5.3 Release Package (APK/AAB + release notes)

Description: Release-ready binaries and notes.

Assigned: DevOps / Release Manager

Expected Output: Signed APK/AAB + release notes

Estimated Timeline: 2–4 days

Tools/Tech: Android Studio, Gradle, React Native CLI, Keystore (app signing)

5.4 Production Deployment & Smoke Test Package

Description: Go-live deployment and verification checklist.

Assigned: DevOps + QA

Expected Output: Deployment record + smoke test result

Estimated Timeline: 1–2 days

Tools/Tech: Android device/emulator, checklist

5.5 Rollback/Hotfix Package

Description: Rollback plan and quick hotfix procedure.

Assigned: DevOps

Expected Output: Rollback steps + tested procedure (if possible)

Estimated Timeline: 1–2 days

Tools/Tech: Git/GitHub, previous APK/AAB

6.0 Documentation

6.1 User Manual Package

Description: Guide for users (setup goals, log workouts, meal plan, view progress).

Assigned: Technical Writer / BA

Expected Output: User manual PDF/Doc

Estimated Timeline: 1 week

Tools/Tech: Google Docs, screenshots

6.2 Admin/Operations Guide Package

Description: Admin/ops procedures (monitoring, troubleshooting, configs).

Assigned: DevOps + Tech Writer

Expected Output: Operations guide

Estimated Timeline: 3–5 days

Tools/Tech: Google Docs

6.3 Technical Documentation Package (Architecture/Firebase/Database)

Description: As-built technical documentation for maintenance.

Assigned: Tech Lead

Expected Output: Architecture notes + Firebase setup/configuration + Firestore data model notes

Estimated Timeline: 1 week

Tools/Tech: Google Docs

6.4 Training & Demo Materials Package

Description: Slides, demo script, and sample scenarios for presentation.

Assigned: BA / Team

Expected Output: PPT + demo checklist

Estimated Timeline: 3–5 days

Tools/Tech: PowerPoint

Requirement Scope Statement

Project Title: Fitness Tracking and Meal Planning Mobile Application

Platform: Android (React Native CLI)

Backend: Firebase (Authentication, Cloud Firestore, FCM)

Project Purpose / Background

The project aims to develop an Android mobile application that helps users track fitness activities and plan meals. The system will store user data securely using Firebase services and provide progress visibility through dashboards and summaries.

Project Objectives

- Develop an Android mobile application that supports users in tracking fitness activities and planning meals.
- Enable users to set goals, record workouts and meals, and view progress through summaries and dashboards.
- Secure user access using Firebase Authentication and store user data reliably in Cloud Firestore.
- Provide reminders/notifications using Firebase Cloud Messaging (FCM) (if enabled in the final build).

Scope Description (High-Level)

The application will include user authentication, profile and goal setup, fitness logging, workout planning, meal planning, nutrition/macro tracking, progress dashboard, notifications, and settings. Data will be stored and managed using Firebase services.

In Scope (Included Requirements)

Functional Scope

- User registration and login using Firebase Authentication
- User profile management (basic info)
- Goal setting (weight goal/activity goal)
- Fitness tracking (manual logging of activities and summaries)
- Workout plan viewing and completion tracking
- Meal planning (daily/weekly meal plan creation and editing)
- Nutrition & macro tracking (food logging and macro totals)
- Progress dashboard and analytics (charts/summaries)
- Push notifications and reminders using Firebase Cloud Messaging (FCM)
- Settings and preferences (e.g., units, notifications, logout)
- Integration items (if implemented per WBS)
 - Google Fit data integration (Android)
 - USDA FoodData Central API for food/nutrition data lookup

Non-Functional Scope (Summary)

- Basic security and privacy controls using Firestore Security Rules
- Data consistency and reliable saving/loading from Cloud Firestore
- Usable UI for Android devices
- Basic performance checks for common screens and Firestore queries

Out of Scope (Excluded Requirements)

- iOS application version
- Payment features and subscriptions
- Personalized coaching by professionals/teleconsultation features
- Advanced AI meal recommendations
- Full wearable ecosystem support beyond Google Fit
- Enterprise-grade analytics, big data pipelines, or external BI tools
- Offline-first mode (unless explicitly required later)

Major Deliverables

- Functional Requirements Document (FRD)
- Non-Functional Requirements Document (NFRD)
- User Stories Backlog
- UX user flows and UI prototype (Figma)
- System architecture documentation
- Firebase data model and security rules specification
- Working Android mobile application (APK/AAB)
- Test plan and test reports
- User Manual
- Admin/Operations Guide
- Technical Documentation
- Training & Demo Materials

Assumptions

- Users have internet access for Firebase services.
- Firebase services are available and accessible during development/testing.
- Testing will be done on Android emulator and/or real Android devices.
- Project features will follow the agreed WBS and approved requirements baseline.

Constraints

- Android-only development (React Native CLI).
- Firebase backend (no custom server API required).
- Limited project duration and team resources (capstone setup)

Stakeholder Register

Project Title: Fitness Tracking and Meal Planning Mobile Application

Stakeholder / Role	Type	Interest / Needs	Influence	Expected Contribution / Responsibility
Project Sponsor / Instructor	External	Project compliance, documentation quality, grading requirements	High	Approves output, evaluates deliverables
Product Owner / Client Representative	External	App meets user needs and scope	High	Gives feedback, approves requirements/UAT
End User (Fitness App User)	External	Easy tracking of workouts and meals; clear dashboard	Medium	Provides usability feedback, UAT input
Mobile Developer (React Native)	Internal	Clear requirements and UI specs	High	Builds mobile app modules and UI
Backend Developer (Firebase)	Internal	Clear data model and security rules	High	Configures Firebase Auth/Firestore/FCM
UI/UX Designer	Internal	Clear user flows and design requirements	Medium	Produces wireframes/prototype/style guide
QA / Tester	Internal	Testable requirements and stable builds	Medium	Creates test plan/cases and runs tests
DevOps/Release (if assigned)	Internal	Smooth release build and deployment process	Medium	Generates APK/AAB, handles release setup
Nutrition Data Provider (USDA FoodData Central API)	External	API usage within limits/terms	Low	Provides nutrition/food lookup data
Google Fit Platform	External	Integration reliability and permissions	Low	Provides activity data via Google Fit API